

Vivek Chavda

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EDUCATION

National Institute of Technology, Surat (SVNIT)
B.Tech in Electronics & VLSI Engineering — CGPA: 8.39/10

Surat, India
Expected 2028

RESEARCH EXPERIENCE

Summer Research Internship Program (SRIP), Indian Institute of Technology Gandhinagar
Research Intern — Generative AI Based Ligand Binding Pose Refinement from Physics Based Prior May 2026 – Jun 2026

- Built a molecular docking pipeline using AutoDock Vina on the PDBbind core set (2020) and an evaluation framework computing RMSD/D-RMSD against crystal poses for 100+ protein-ligand complexes.
- Studied and executed FLOWR, an open-source flow-matching generative model, and trained it on a 500-complex subset of the SPINDR dataset for physics-informed pose refinement.
- Proposed and validated a hybrid pipeline combining physics-based docking priors with generative flow-matching models to improve binding pose accuracy over either method used independently.

PROJECTS

Hedgefund AI: Autonomous Equity Research Agent | [GitHub](#) | *Python, LangGraph, Llama-3* Jan 2026

- Built a multi-agent LLM system with LangGraph and a RAG pipeline to automate financial research from real-time market data and news.
- Designed structured report generation with reflection-based validation; optimized query workflows reducing latency by 40%.

YouTube Chatbot: AI-Powered RAG Agent | [GitHub](#) | *Python, LangChain, Groq API, FAISS* Mar 2026

- Developed a RAG-based chatbot for conversational Q&A over YouTube videos via transcript extraction and semantic search.
- Implemented MiniLM embeddings with FAISS retrieval and integrated LLaMA-3/Mixtral via Groq into a Streamlit UI.

AI-Powered Land Cover Classification | [GitHub](#) | *Python, PyTorch, GeoPandas, Rasterio* Feb 2026

- Trained a ResNet-18 model on 9,200+ satellite images across 5 land cover classes with a geospatial preprocessing pipeline.
- Automated patch extraction and CRS transformations; achieved 89.45% accuracy with spatial validation.

IPL Match Predictor & Score Forecaster | [GitHub](#) | *Python, XGBoost, Scikit-learn, Streamlit, Pandas* Dec 2025

- Built XGBoost models for match prediction (80% accuracy) and score forecasting ($R^2 = 0.7$) with venue bias and run-rate features.
- Deployed an interactive forecasting interface using Streamlit.

PUBLICATIONS

MELD-SR: Multi-Expert Learned Decomposition Using Frequency Guidance | *NTIRE Workshop, CVPRW 2026 (PyTorch)*

- Co-authored paper accepted at the NTIRE Workshop, CVPRW 2026; designed a frequency-guided fusion network (~1M parameters) achieving 31.48 dB PSNR, ranked 20th globally on the restoration track.

TECHNICAL SKILLS

Languages: Python, Java

AI/ML & LLMs: PyTorch, LangChain, LangGraph, XGBoost, Scikit-learn, FAISS, Groq API, HuggingFace, Computer Vision, RAG, Generative AI, Diffusion Models, Flow Matching

Libraries & Frameworks: FastAPI, Streamlit, Pandas, NumPy, GeoPandas, Rasterio

Tools & Deployment: Git, GitHub, Docker

Fundamentals: Data Structures & Algorithms, OOP

CERTIFICATIONS

Supervised Machine Learning | DeepLearning.AI

Advanced Learning Algorithms | DeepLearning.AI